FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2024

TEST DATE: 20-01-2024

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The driver of a train moving at a speed 'v_1' sees a goods train a distance 'd' ahead of him on the same track and moving in the same direction with a slower speed 'v_2'. He puts on breaks and gives his train a constant retardation 'a'. There will be no collision if:

(A) $d < \frac{(v_1+v_2)^2}{a}$ (B) $d > \frac{(v_1-v_2)^2}{2a}$

(C) $d > \frac{(v_1-v_2)^2}{a}$ (D) $d < \frac{(v_1-v_2)^2}{a}$

2. A simple pendulum with length *L* and mass *m* (of the bob) is vibrating with an amplitude *a*. The maximum tension in the string is

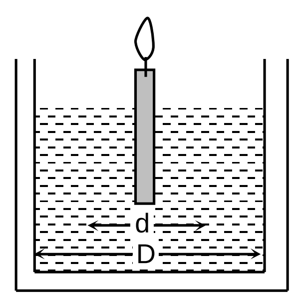
(A) *mg* (B) $mg\left[1+\left(\frac{a}{L}\right)^2\right]$

(C) $\left[1+\frac{a}{2L}\right]^2$ (D) $mg\left[1+\left(\frac{a}{L}\right)\right]^2$

3. Electromagnetic radiation whose electric component varies with time as $E = C_1(C_2 + C_3\cos\omega t)\cos\omega_0 t$, here C_1, C_2 and C_3 are constants, is incident on lithium and librates photoelectrons. If the kinetic energy of most energetic electrons be 0.592 × 10^{-19} J. Given that ω_0 = 3.6 × 10^{15} rad/sec and ω = 6 × 10^{14} rad/sec. The work function of lithium is (take planks constant *h* = 6.6 × 10^{-34} MKS).

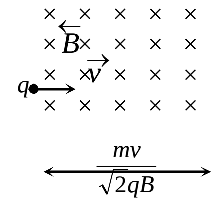
(A) 1.2 eV (B) 1.5 eV
(C) 2.1 eV (D) 2.39 eV

4. A candle of diameter d is floating on a liquid in a cylindrical container of diameter D (D >> d) as shown in figure. If it is burning at the rate of 2 cm/hour then the top of the candle will (assume) that density of wax is half the density of the liquid



(A) remain at the same height
(B) fall at the rate of 1 cm/hour
(C) fall at the rate of 2 cm/hour
(D) go up at the rate of 1 cm/hour

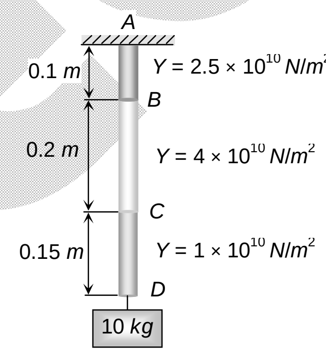
5. A positive charge q is projected in magnetic field of width $\frac{mv}{\sqrt{2}\,qB}$ with velocity v as shown in figure. Then time taken by charged particle to emerge from the magnetic field is



(A) $\frac{m}{\sqrt{2}\,qB}$
(B) $\frac{\pi m}{4qB}$
(C) $\frac{\pi m}{2qB}$
(D) $\frac{\pi m}{\sqrt{2}\,qB}$

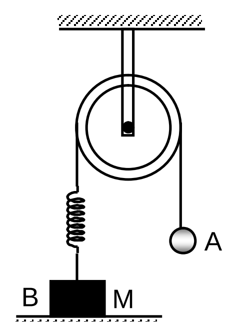
6. For same braking force the stopping distance of a vehicle increases from 15 m to 60 m. By what factor the velocity of vehicle has been changed
(A) 2
(B) 3
(C) 4
(D) $3\sqrt{5}$

7. A light rod with uniform cross-section of $10^{-4}\,m^2$ is shown in the adjoining figure. The rod consists of three different materials whose lengths are 0.1 *m*, 0.2 *m* and 0.15 *m* respectively and whose Young's modulii are $2.5\times10^{10}\ N/m^2$, $4\times10^{10}\ N/m^2$ and $1\times10^{10}\ N/m^2$ respectively. The displacement of point *B* will be



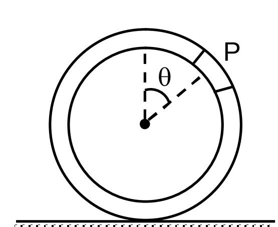
(A) $24\times10^{-6}m$
(B) $9\times10^{-6}m$
(C) $4\times10^{-6}m$
(D) $1\times10^{-6}m$

8. In the adjoining diagram, the ball A is released from rest when the spring is at its natural length (neither stretched nor compressed). For the block B of mass M to leave contact with the ground at some time, the minimum mass of A must be:



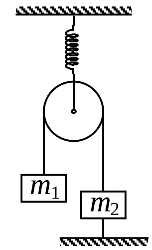
(A) $\frac{M}{2}$
(B) M
(C) 2M
(D) a function of M and force constant k of spring

9. A small block of mass m is rigidly attached to P to a ring of mass 3 m and radius r. The system is released from rest at $\theta = 90°$ and rolls without sliding. The angular acceleration of hoop after release is:



(A) $\frac{g}{4r}$
(B) $\frac{g}{8r}$
(C) $\frac{g}{3r}$
(D) $\frac{g}{2r}$

10. In the figure shown, pulley and spring are ideal. If k is spring constant of spring, the potential energy stored in it is ($m_1 > m_2$)



(A) $\frac{2m_1^2 g^2}{k}$ (B) $\frac{2m_2^2 g^2}{k}$

(C) $\frac{(m_1+m_2)^2 g^2}{k}$ (D) $\frac{1}{2}\frac{(m_1-m_2)^2 g^2}{k}$

11. A thin uniform rod of mass 1 kg and length 2 m is free to rotate about its upper end. When it is at rest, it receives an impulse of 10 Ns at its lowest point, normal to its length. The value of angular velocity of rod just after impact is

(A) 10 rad/s (B) 15 rad/s
(C) 20 rad/s (D) 25 rad/s

12. A wire is of mass (0.3 $\pm$ 0.003) gm. The radius is (0.5 $\pm$ 0.005) mm and length is (6.0 $\pm$ 0.06) cm then % error in density is

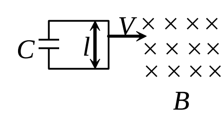
(A) 3 (B) 4
(C) 6 (D) –2

13. A projectile is fired from the surface of earth of radius R with a velocity kv_e where v_e is the escape velocity and $k<1$. Neglecting air resistance, the maximum height of rise will be

(A) $\frac{R}{k^2-1}$ (B) k^2R

(C) $\frac{k^2R}{1-k^2}$ (D) kR

14. A capacitance C is connected to a conducting rod of length l moving with a velocity v in a transverse magnetic field B then the charge developed in the capacitor is



(A) zero (B) BlvC

(C) $\frac{BlvC}{2}$ (D) $\frac{BlvC}{3}$

15. Temperature of source is 330ºC. Temperature of sink is changed in order to increase the efficiency of engine from $\frac{1}{5}$ to $\frac{1}{4}$, by

(A) 30ºK (B) 303ºK
(C) 603ºK (D) 60ºK

16. A particle moves on a rough horizontal ground with some initial velocity say v_0. If 3/4th of its kinetic energy is lost in friction in time t_0. Then co-efficient of friction between the particle and ground is

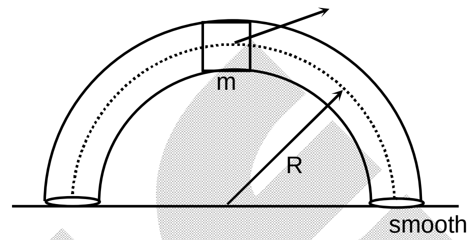
(A) $\frac{v_0}{2gt_0}$ (B) $\frac{v_0}{4gt_0}$

(C) $\frac{3v_0}{4gt_0}$ (D) $\frac{3v_0}{gt_0}$

17. The potential energy of a particle of mass 1 kg in motion along the *x*-axis is given by $U = 4(1-\cos 2x)J$. Here *x* is in metres. The period of small oscillations (in sec) is

(A) 2π
(B) π
(C) $\frac{\pi}{2}$
(D) $\sqrt{2}\ x$

18. In a vertical plate inside a smooth hollow thin tube a block of same mass as that of tube is released as shown in figure. When it is slightly disturbed it moves towards right. By the time the block reaches the right of the tube them the displacement of the tube will be [R is the mean radius of tube. Assume that the tube remains in vertical plane]:



(A) $2R/\pi$
(B) $4R/\pi$
(C) R/2
(D) R

19. A wooden block of mass *M* rests on a horizontal surface. A bullet of mass *m* moving in the horizontal direction strikes and gets embedded in it. The combined system covers a distance *x* on the surface. If the coefficient of friction between wood and the surface is μ, the speed of the bullet at the time of striking the block is (where *m* is mass of the bullet)

(A) $\sqrt{\frac{2Mg}{\mu m}}$
(B) $\sqrt{\frac{2\mu mg}{Mx}}$
(C) $\sqrt{2\mu gx}\left(\frac{M+m}{m}\right)$
(D) $\sqrt{\frac{2\mu mx}{M+m}}$

20. If the radius of the earth were to shrink by one per cent, its mass remaining the same, the value of g on the earth's surface would

(A) increase by 0.5%
(B) increase by 2%
(C) decrease by 0.5%
(D) decrease by 2%

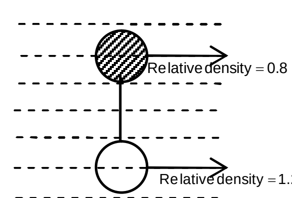
SECTION – B

(Numerical Answer Type)

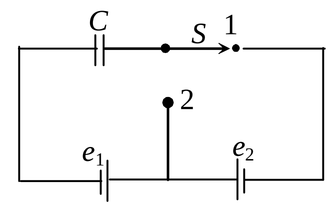
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. Two metallic strings A and B of different materials are connected in series forming a joint. The strings have similar cross-sectional area. The length of A is l_A = 0.3 and that of B is l_B = 0.75 m. One end of the combined string is tied with a support rigidly and the other end is loaded with a block of mass m passing over a frictionless pulley. Transverse waves are set up in the combined string using an external source of variable frequency. Calculate (i) the lowest frequency for which standing waves are observed such that the joint is a node and (ii) the total number of antinodes at this frequency. The densities of A and B are 6.3×10^3 kg m^{-3} and 2.8×10^{-3} kg m^{-2} respectively.

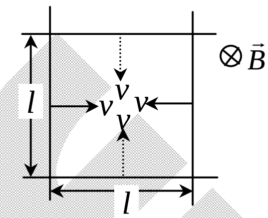
22. Two spheres of volume 500 cc each but of relative densities 0.8 and 1.2 are connected by the string and the combination is immersed in a liquid in vertical position as in figure. Find the tension in the string (in Newton). (g = 10 m/s^2)



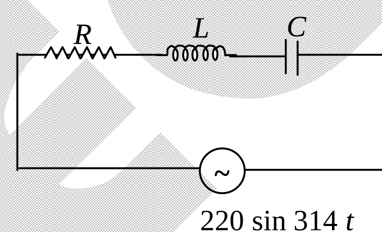
23. What amount of heat in μJ will be generated in the circuit shown in the figure after the switch is shifted from position 1 to position 2? (given that C = 2 μF, e_1 = 20 V, e_2 = 10 V)



24. In the figure shown the four rods have λ = 0.5 Ω/m resistance per unit length. The arrangement is kept in a magnetic field of constant magnitude *B* = 2T and directed perpendicular to the plane of the figure and directed inwards. Initially the rods form a square of side length *l* = 15 m as shown. Now each wire starts moving with constant velocity *v* = 5 m/s towards opposite wire. Find the force required in newton on each wire to keep its velocity constant at *t* = 1.



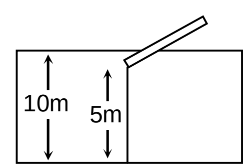
25. In the circuit shown, the value of *L* is 5 henry and the power factor of the circuit is 0.8. It is also given that the voltage drop across capacitor is $\frac{2}{5}$ times the voltage drop across the inductor. Find impendence (in ohm) of the circuit.



26. An α-particle is accelerated by a potential difference of 10^4V. Find the change in its direction (in degree) of motion as angle measured in degree if it enters normally in a region of thickness 0.1 m having transverse magnetic induction of 0.1 *T*
[given mass of α-particle = 6.4×10^{-27} kg].

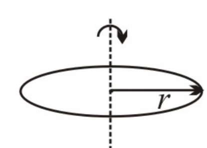
27. An expansible balloon filled with 2/3 of its volume submerged. How deep should it be sunk in water, in order that it is just in equilibrium, neither sinking nor rising further? The height of water barometer = 10 m.

28. A rod of length 6 m has specific gravity $\rho(=25/36)$. One end *c* the rod is tied to 5m long rope. Which in turn is tied to the floor of a pool 10 m deep as shown. Find the length of the part of rod which is out of water.



29. An isolated and charged spherical soap bubble has a radius r and the pressure inside is atmospheric. If T is the surface tension of soap solution, then charge on drop is $X\pi r\sqrt{2rT\varepsilon_0}$. Find the value of X.

30. A ring of radius *r* made of wire of density ρ is rotated about a stationary vertical axis passing through its centre and perpendicular to the plane of the ring as shown in figure. Determine the angular velocity (in rad /s) of ring at which the ring breaks. The wire breaks at tensile stress σ. Ignore gravity. (Take $\frac{\sigma}{\rho}$ = 4 and *r* = 1 m)



Chemistry

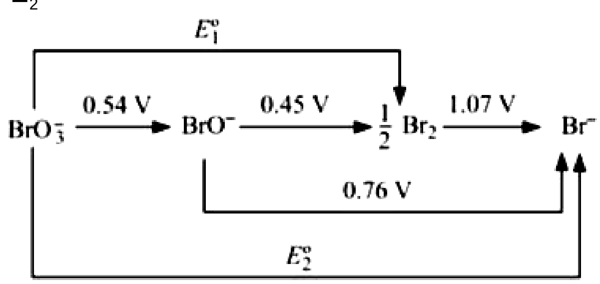
PART – B

SECTION – A
(One Options Correct Type)

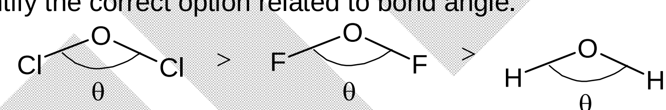
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

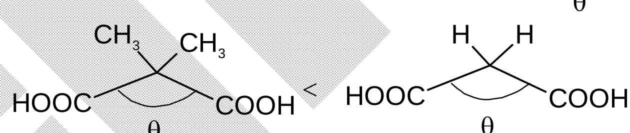
31. In a measurement of quantum efficiency of photosynthesis in green plants, it was found that 10 quanta of red light of wavelength 6850Å were needed to release one molecule of O_2. The average energy storage in this process for 1 mole O_2 evolved is 112 kcal. What is the energy conversion efficiency in this experiment (in percentage).
Given: 1 cal = 4.18 J; $N_A = 6 \times 10^{23}$; $h = 6.63 \times 10^{-34}$ J.s
(A) 23.5 (B) 26.9
(C) 66.34 (D) 73.1

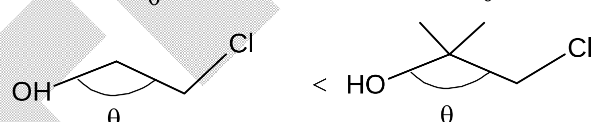
32. From the standard potential shown in the following diagram, calculate the potentials E_1° and E_2°



(A) $E_1^\circ = 0.52\,V, E_2^\circ = 0.61\,V$ (B) $E_1^\circ = 0.52\,V, E_2^\circ = 0.52\,V$
(C) $E_1^\circ = 0.61\,V, E_2^\circ = 0.79\,V$ (D) $E_1^\circ = 0.44\,V, E_2^\circ = 0.88\,V$

33. Identify the correct option related to bond angle.
(A)


(B)


(C)


(D) $PH_3 > NH_3 > OCl_2$

34. A sample of camphor used in the Rast method for the determination molar mass had melting point of 451.55 K. The melting point of a solution containing 0.522 g camphor and 0.038.6 g of an unknown compound was 433.85 K. If the unknown compound was hydrocarbon containing 7.7% H, find its molecular formula. The molar mass of camphor is 152.2 g mol^{-1} and its $\Delta_{fus}H_{1,\,m}$ = 6.844 kJ mol^{-1}
(A) C_6H_6 (B) $C_{12}H_{12}$
(C) $C_{14}H_{14}$ (D) C_8H_{16}

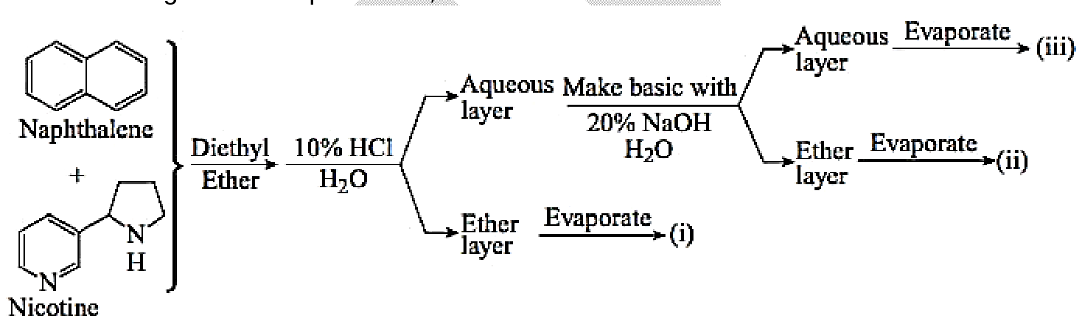
35. For the reaction
$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$
The equilibrium pressure at 298 K was found to be 0.67 bar. Calculate and Δ_rG^o of the reaction.
(A) 5.42 kJ mol^{-1}
(B) 10.84 kJ mol^{-1}
(C) 1.42 kJ mol^{-1}
(D) 4.42 kJ mol^{-1}

36. In each of the following reactions, role of water is:
(i) $H_2O + HCl \rightarrow H_3O^+ + Cl^-$
(ii) $6H_2O + Mg^{2+} \rightarrow [Mg(H_2O)_6]^{2+}$
(iii) $2H_2O + 2F_2 \rightarrow 4\ HF + O_2$
(A) (i) oxidant; (ii) reductant; (iii) base
(B) (i) reductant; (ii) oxidant; (iii) base
(C) (i) base; (ii) base; (iii) reductant
(D) (i) acid; (ii) base; (iii) reductant

37. A standard 0.0100 M solution of Na^+ is required to calibrate an ion selective electrode. Describe how this solution can be prepare from primary-standard Na_2CO_3 (105.99 g/mol)
(A) By dissolving 265 g of Na_2CO_3 in 500 ml solution
(B) By dissolving 265 g of Na_2CO_3 in 1000 ml solution
(C) By dissolving 265 mg of Na_2CO_3 in water and diluting to 500 ml by water
(D) By dissolving 265 mg of Na_2CO_3 in water and diluting to 1000 ml by water.

38. Calculate the solubility of $Ba(IO_3)_2$ in a solution prepared by mixing 200 ml of 0.0100 M $Ba(NO_3)_2$ with 100 ml of 0.100 M $NaIO_3$ ($K_{sp} = 1.57 \times 10^{-9}$)
(A) $3.9. \times 10^{-6}$ M
(B) 4.93×10^{-5}
(C) 3.93×10^{-3} M
(D) None of these

39. In the following extraction procedure, choose the number where nicotine would be found.



(A) i = Nicotine
(B) i + iii = Nictone
(C) ii = Nicotine
(D) ii + iii = Nictone

40. In a solution as well as in the solid state, the hydroxide of sodium, potassium and barium react rapidly with atmospheric carbon dioxide produce the corresponding carbonate, shown in given reaction
$$\underset{(g)}{CO_2} + 2OH^-(aq) \rightarrow \underset{(aq)}{CO_3^{2-}} + H_2O$$
During standardization of NaOH (aq) solution which of the following statement is correct
(A) Absorption of CO_2 by aqueous NaOH solution leads to error in the analysis when the acid range indicator is used
(B) Absorption of CO_2 by aqueous NaOH solution leads to error in the analysis when the basic range indicator is used is called carbonate error.
(C) Phenolphthalein is acid range indicator
(D) Phenolphthalein indicator can be used to standardize NaOH aqueous solution containing CO_2 gas.

41. When one CO group is replaced by PPh_3 in $[Cr(CO)_6]$, which one of the following statements is TRUE?
(A) The Cr-C bond length increases and CO bond length decreases
(B) The Cr-C bond length decreases and CO bond length decreases
(C) The Cr-C bond length decreases and CO bond length increases
(D) The Cr-C bond length increases and CO bond length increases

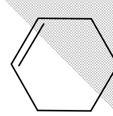
42. The correct order of energy levels of the molecular orbitals of N_2 is
(A) $1\sigma_g < 1\sigma_u < 2\sigma_g < 2\sigma_u < 1\pi_u < 3\sigma_g < 1\pi_g < 3\sigma_u$
(B) $1\sigma_g < 1\sigma_u < 2\sigma_g < 2\sigma_u < 3\sigma_g < 3\sigma_u < 1\pi_u < 1\sigma_g$
(C) $1\sigma_g < 1\sigma_u < 2\sigma_g < 2\sigma_u < 1\pi_g < 3\sigma_g < 1\pi_u < 3\sigma_u$
(D) $1\sigma_g < 1\sigma_u < 2\sigma_g < 2\sigma_u < 3\sigma_g < 1\pi_g < 1\pi_g < 3\sigma_u$

43. In trigonal bipyramidal there occur a geometrical isomer.

: Compound:	Geometrical isomers
• $Fe(CO)_4(PPh_3)$	x
• EX_2Y_3	y

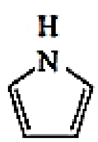
(E-central atom with 5σ bond, zero lone pair)
Identify the correct option related to geometrical isomers
(A) x = 2, y = 3 (B) x = 1, y = 3
(C) x = 3, y = 3 (D) x – 1, y = 2

44. If enthalpy of hydrogenation of $C_6H_6(l)$ into $C_6H_{12}(l)$ is –205 kJ and resonance energy of $C_6H_6(l)$ is –152

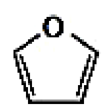
kJ/mol then enthalpy of hydrogenation of  is

Assume ΔH_{vap} of $C_6H_6(l)$, $C_6H_8(l)$, $C_6H_{12}(l)$ all are equal
(A) –535.5 kJ/mol (B) –238 kJ/mol
(C) –357 kJ/mol (D) –119 kJ/mol

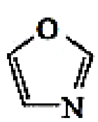
45. Which of the following compounds are aromatic compounds?

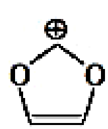
(i) 

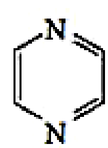
(ii) 

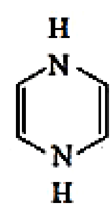
(iii) 

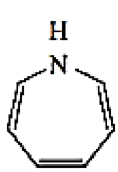
(iv) 

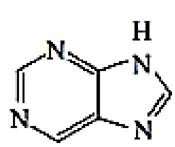
(v) 

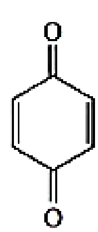
(vi) 

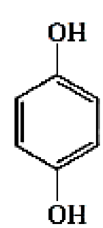
(vii) 

(viii) 

(ix) 

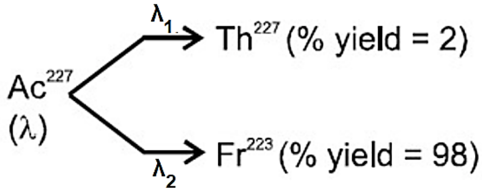
(x) 

(xi) 

(xii) 

(A) i, ii, iii, iv, v, vi, viii, x, xii (B) i, ii, iii, vi, viii, x, xi, xi, xii
(C) i, ii, iii, iv, v, vi, vii, x, xii (D) i,ii, iii, v, vii, viii, ix, x, xii

46. Ac^{227} has a half-life of 22 years. The decays follow two parallel paths

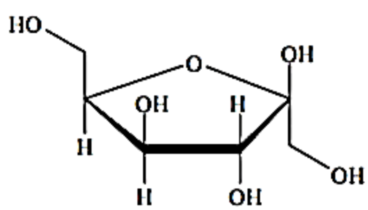


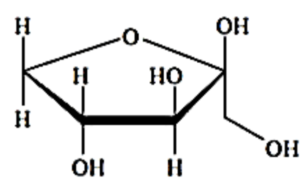
What are the decay constant (λ) for Th and Fr respectively?
(A) 0.03087, 0.00063
(B) 0.00063, 0.03087
(C) 0.02, 0.98
(D) None of these

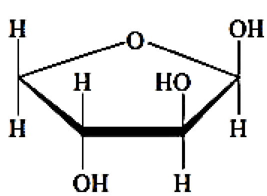
47. A, B and C are hydroxy-compounds of the elements X, Y and Z respectively. X, Y and Z are in the same period of the periodic table. A gives an aqueous solution of pH less than seven. B reacts with both strong acid and strong alkalis. C gives an aqueous solution which is strongly alkaline.
I. The three elements are metals.
II: The electronegativities decrease from X to Y to Z.
III: The atomic radius decreases in the order X, Y and Z
IV: X, Y and Z could be phosphorus, aluminium and sodium respectively.
(A) I, II and III only correct
(B) I, III only correct
(C) II, IV only correct
(D) II, III, IV only correct

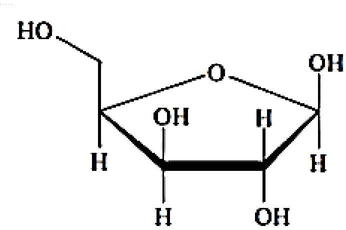
48. Incorrect order of ionic size is:
(A) $La^{3+} > Gd^{3+} > Eu^{3+} > Lu^{3+}$
(B) $V^{2+} > V^{3+} > V^{5+} > Mn^{7+}$
(C) $Tl^{+} > In^{+} > Sn^{2+} > Sb^{3+}$
(D) $K^{+} > Sc^{+} > V^{5+} > Mn^{7+}$

49. Which of the following compounds is a β-aldopentafuranose?

(i)


(ii)


(iii)


(iv)


(A) i
(B) ii
(C) iii
(D) iv

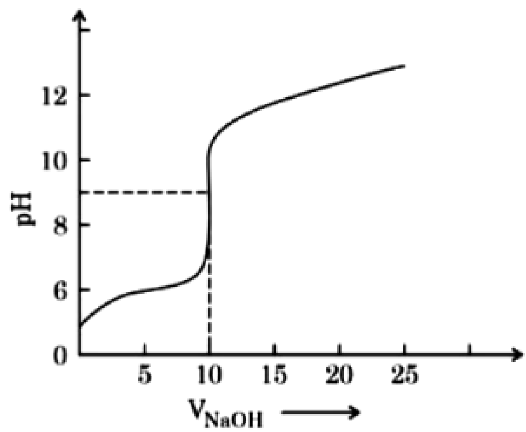
50. A six coordinate complex of formula $CrCl_3.6H_2O$ has green colour. A 0.1 M solution of the complex when treated with excess of $AgNO_3$ gave 28.7g of white precipitate. The formula of the complex would be
(A) $[Cr(H_2O)_6]Cl_3$
(B) $[CrCl(H_2O)_5]Cl_2.\ H_2O$
(C) $[CrCl_2(H_2O)_4]Cl.\ 2H_2O$
(D) $[Cr(H_2O)_3Cl_3]$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. Given below is the plot of pH vs volume of NaOH added in the acid-base titration. The correct number of statemen/s among the following
(a) Before the equivalence point, a series of buffer solutions determine the pH.
(b) The graph represents the titration of a strong acid with NaOH
(c) At the equivalence point, hydrolysis of the anion of the acid determines the pH
(d) After the equivalence point acid/salt buffer solution determines the pH.

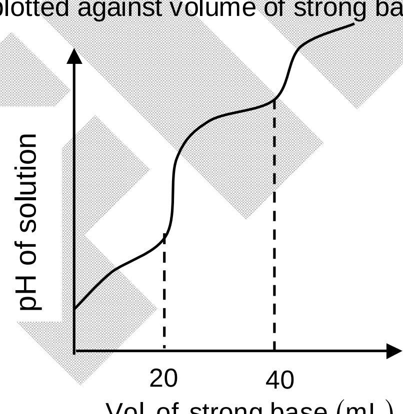


52. Carbon monoxide reacts with O_2 to form CO_2: $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$ information on this reaction is given in the table below.

[CO] mol/L	[O_2] mol/L	Rate of reaction (mol/L. min)
0.02	0.02	4×10^{-5}
0.04	0.02	1.6×10^{-4}
0.02	0.04	8×10^{-5}

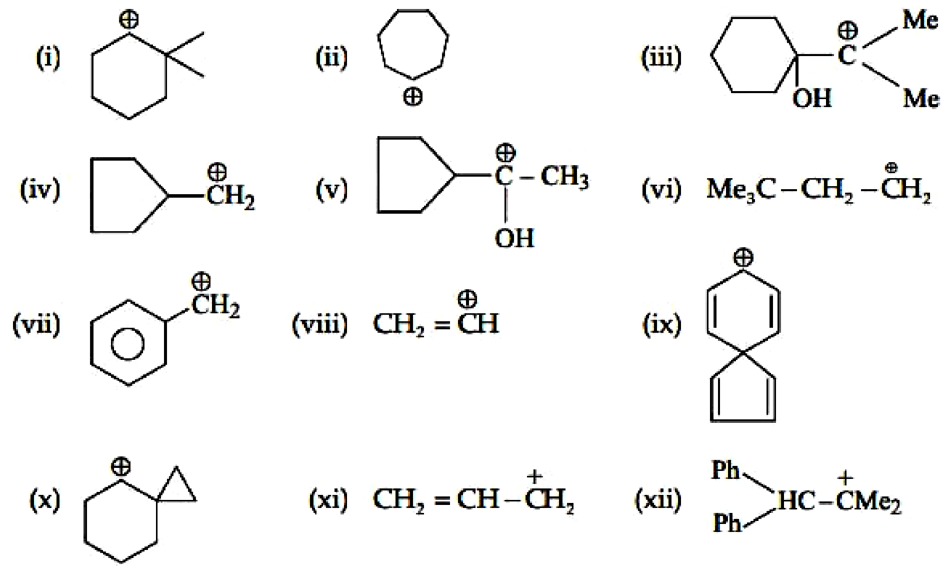
What is the value for the rate of constant for the reaction in proper related unit?

53. 10 mL of H_2A (weak diprotic acid) solution is titrated against 0.1M NaOH. pH of the solution is plotted against volume of strong base added and following observation is made.



If pH of the solution at first equivalence point is pH_1 and at second equivalence point is pH_2. Calculate the value of $(pH_2 - pH_1)$ at 25ºC
Given: For H_2A, $pK_{a_1} = 4.6$ and $pK_{a_2} = 8$, log 25 = 1.4

54. How many carbocations undergo rearrangements?



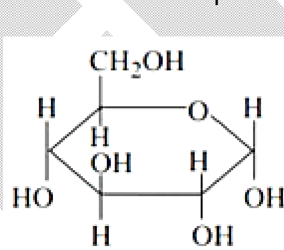
55. Identify the number of correct statement
(a) Mobility of the alkali metal ions in aqueous solution following the sequence $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
(b) When LiI and NaF heated we get LiF and NaI
(c) Enthalpy of hydration of Na^+, $K^{\oplus}$ and Rb^+ is in values of $Na^+ > K^+ > Rb^+$
(d) $Li_3N + H_2O \rightarrow LiOH + (A)$, compound is NH_3

56. Molar conductivities at infinite dilution of KCl, HCl and CH_3COOK are 0.013, 0.038 and 0.009 Sm^2mol^{-1} respectively at 291 K. If conductivity of 0.001M CH_3COOH is 2.72×10^{-3} Sm^{-1} then find % degree of dissociation of CH_3COOH.

57. If enthalpy of neutralisation of HCl by NaOH is –57 kJ mol^{-1} and with NH_4OH is –50 kJ mol^{-1}. Calculate enthalpy of ionisation of $NH_4OH(aq)$.

58. 10 g mixture of $K_2Cr_2O_7$ and $KMnO_4$ was treated with excess of KI in acidic medium. Iodine liberated required 100 cm^3 of 2.2 N sodium thiosulphate solution for titration. If the mass percent of $KMnO_4$ in the mixture Z, then what is the value of 2Z/5?

59. The number of unpaired electron in $[Co(H_2O)_6]^{2+}$ is_______.

60.  $\xrightarrow[H_2O]{Br_2}$ Compound (A); Number of chiral centre in compound A is

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $f''(x) > 0$, $\forall x \in R$, $f'(3) = 0$ and $g(x) = f(\tan^2 x - 2\tan x + 4)$, $0 < x < \frac{\pi}{2}$, then g (x) is increasing in

(A) $\left(0, \frac{\pi}{4}\right)$
(B) $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$
(C) $\left(0, \frac{\pi}{3}\right)$
(D) none of these

62. The area of the region whose boundaries are defined by the curves y = 2 cos x, y = 3 tan x and the y-axis, is

(A) $1+3\ln\left(\frac{2}{\sqrt{3}}\right)$
(B) $1 + \frac{3}{2}\ln 3 - 3\ln 2$
(C) $1 + \frac{3}{2}\ln 3 - \ln 2$
(D) $\ln 3 - \ln 2$

63. Let A, B, C be 3 independent events such that $p(A) = \frac{1}{3}$, $p(B) = \frac{1}{2}$, $p(C) = \frac{1}{4}$. Then probability of exactly 2 events occurring out of 3 events is ______________________

(A) 1/4
(B) 1/2
(C) 1/3
(D) 1/16

64. The centre of the circle obtained by reflecting the circle $x^2 + y^2 = m^2 + m + 1$ in the line y = mx + m, is

(A) $\left(-\frac{2m^2}{1+m^2}, -\frac{2m}{1+m^2}\right)$
(B) $\left(-\frac{2m^2}{1+m^2}, \frac{2m}{1+m^2}\right)$
(C) $\left(\frac{2m}{1+m^2}, \frac{2m^2}{1+m^2}\right)$
(D) $\left(\frac{2m}{1+m^2}, -\frac{2m^2}{1+m^2}\right)$

65. The vertices of a triangle in the argand plane are 3 + 4i, 4+ 3i and $2\sqrt{6}$ + i, then distance between orthocentre and circumcentre of the triangle is equal to,

(A) $\sqrt{137-28\sqrt{6}}$
(B) $\sqrt{137+28\sqrt{6}}$
(C) $\frac{1}{2}\sqrt{137+28\sqrt{6}}$
(D) $\frac{1}{3}\sqrt{137+28\sqrt{6}}$

66. If x, y, z are in G.P (x, y, z > 0), then the determinant $\begin{vmatrix} xp+y & x & y \\ py+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix}$ is equal to

(A) 1
(B) 0
(C) a multiple of p
(D) None of these

67. f(x) = $\begin{cases} 3[x]-5\dfrac{|x|}{x}, & x \neq 0 \\ 2, & x = 0 \end{cases}$, then $\int_{-3/2}^{2} f(x)dx$ is equal to ([.] denotes the greatest integer function)

(A) $-\dfrac{11}{2}$
(B) $-\dfrac{7}{2}$
(C) –6
(D) $-\dfrac{17}{2}$

68. Let f(x) = $(x^2 - 4)|(x^3 - 6x^2 + 11x - 6)| + \dfrac{x}{1+|x|}$. The set of points at which the function f(x) is not differentiable is

(A) {–2, 2, 1, 3}
(B) {–2, 0, 3}
(C) {–2, 2, 0}
(D) {1, 3}

69. The number of solutions of the equation $x^2 - 3[\sin x] = 3$ (where [.] denotes the greatest integer function) is

(A) one
(B) two
(C) four
(D) six

70. The length of the major axis of the ellipse

$$(5x-10)^2 + (5y+15)^2 = \frac{(3x-4y+7)^2}{4}$$ is

(A) 10
(B) $\dfrac{20}{3}$
(C) $\dfrac{20}{7}$
(D) 4

71. Let f (x) be defined for all x $\in$ R and be continuous. Let f (x + y) – f (x – y) = 4xy $\forall$ x, y $\in$ R and f (0) = 0 then

(A) f (x) is bounded
(B) $f(x) + f\left(\dfrac{1}{x}\right) = f\left(x + \dfrac{1}{x}\right) + 2$
(C) $f(x) + f\left(\dfrac{1}{x}\right) = f\left(x - \dfrac{1}{x}\right) + 2$
(D) none of these

72. If ω is the complex cube root of unity, then $\begin{bmatrix} \omega^2 & -\omega^2 \\ \omega & \omega \end{bmatrix}^{-1} \begin{bmatrix} \omega & \omega^2 \\ -\omega & \omega^2 \end{bmatrix}^{-1}$ is equal to

(A) $\begin{bmatrix} \omega & \omega^2 \\ \omega & \omega \end{bmatrix}$
(B) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
(C) $\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$
(D) none of these

73. $\vec{OA}$ and $\vec{OB}$ are two vectors such that $|\vec{OA}+\vec{OB}| = |\vec{OA}+2\vec{OB}|$. Then ;

(A) $\angle BOA = 90°$
(B) $\angle BOA > 90°$
(C) $\angle BOA < 90°$
(D) $60° \le \angle BOA \le 90°$

74. The incentre of the triangle formed by the lines y = |x| and y = 1 is

(A) $(0, 2 - \sqrt{2})$
(B) $(2 - \sqrt{2}, 0)$
(C) $(2 + \sqrt{2}, 0)$
(D) $(0, 2 + \sqrt{2})$

75. For n $\in$ N, let $f_n(x) = \tan\frac{x}{2}(1+\sec x)(1+\sec 2x)(1+\sec 4x).......(1+\sec 2^n x)$. Then $\lim\limits_{x\to 0}\frac{f_n(x)}{2x}$ is equal to

(A) 0
(B) 2^n
(C) 2^{n-1}
(D) 2^{n+1}

76. The value of $\lim\limits_{x\to\infty} x\left[\tan^{-1}\left(\frac{2x+1}{2x+4}\right)-\frac{\pi}{4}\right]$ is equal to

(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) $\frac{3}{4}$
(D) $-\frac{3}{4}$

77. If one end of the diameter of a circle is (3, 4) which touches the x-axis then the locus of other end of the diameter of the circle is

(A) parabola
(B) hyperbola
(C) ellipse
(D) none of thee

78. The complete set of values of x satisfying $\log_x (x^2 - 2) \ge 0$ is

(A) $0 < x \le \sqrt{3}$
(B) $\sqrt{2} < x \le \sqrt{3}$
(C) $x \ge \sqrt{3}$
(D) none of these

79. The coordinates of the point on the parabola $y = x^2 + 7x + 2$, which is nearest to the straight line $y = 3x - 3$ are

(A) (–2, –8)
(B) (1, 10)
(C) (2, 20)
(D) (–1, –4)

80. If lines x = y = z, $x = \frac{y}{2} = \frac{z}{3}$ and third line passing through (1, 1, 1) form a triangle of area $\sqrt{6}$ units then point of intersection of third line with second line will be
(A) (1, 2, 3)
(B) (2, 4, 6)
(C) $\left(\frac{4}{3}, \frac{8}{3}, \frac{12}{3}\right)$
(D) none of these

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. $a^4 + \frac{1}{a^4} = 119$ (a > 0), then value of $a^3 - \frac{1}{a^3}$ is

82. If $\int \frac{(x^2-1)}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)} dx = k \log\left|\tan^{-1}\frac{x^2+1}{x}\right| + c$, then k is equal to

83. The least integral value of k such that $(k-2)x^2 + 8x + k + 4$ is positive for all real values of x is

84. Number of solutions to the equation $\sin^{-1}x - \cos^{-1}x = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is

85. If $\sin\theta + \sin\phi = \sqrt{3}(\cos\phi - \cos\theta)$, then $\sin 3\theta + \sin 3\phi$ is equal to

86. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = x_1\hat{i} + x_2\hat{j} + x_3\hat{k}$, where $x_1, x_2, x_3 \in \{-3, -2, -1, 0, 1, 2\}$.
Number of possible vectors $\vec{b}$ such that $\vec{a}$ and $\vec{b}$ are mutually perpendicular, is

87. If $\tan x - \tan^2 x = 1$, then the value of $\tan^4 x - 2\tan^3 x - \tan^2 x + 2\tan x + 1$ is

88. If a, b, c, d are four consecutive, non-negative integers, such that $a = b^2 - c^2 + d^2$, then $a^2 - b^2 + c^2 - d^2$ is

89. The value of $\sum_{r=1}^{1024} [\log_2 r]$ is equal to, ([.] denotes the greatest integer function)

90. The value of $\int_{-1}^{1} [x[1 + \sin \pi x] + 1]\, dx$ is, ([.] denotes the greatest integer)